Tutorial 3

Randomized strategies and noisy best responses

with step-by-step solutions

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How this tutorial works

- Four blocks. Each one opens with a compact recap of the theory and a pointer to the lecture where it is developed, and then moves to exercises on that same material.
- The exercises use the same templates as the online quizzes on Virtuale and as the mid-term. Same mechanics, different numbers.
- Try each one first. We solve it together afterwards.
- Questions on the *content* of the course go to Prof. Bigoni. Questions on the organisation, the quizzes and the exam come to me: `marco.rosso4@unibo.it`

Coverage: Lectures 9, 10.

Plan

1. Matching pennies: why and how to randomise
2. Battle of the sexes and coordination: mixed equilibria
3. Payoff imbalances and noisy best responses
4. The playing-cards exercise of Lecture 10

Sample *oral* questions at the end.

PART 1

Matching pennies

Mixed strategies: the recipe

Lecture 9 – “Compliance and control”, “Expected payoffs”, “Summing up”

	Control	No control
Compliance	$1, -1$	$-1, 1$
No compliance	$-1, 1$	$1, -1$

No equilibrium in pure strategies: being predictable is exploited.

p = probability of Right (No control):

$$\mathbb{E}[\text{Top}] = 1(1-p) - 1 \cdot p = 1 - 2p$$

$$\mathbb{E}[\text{Bottom}] = -1(1-p) + 1 \cdot p = 2p - 1$$

$$1 - 2p = 2p - 1 \;\Rightarrow\; p = 1/2$$

Step 1. Notation: p = Pr(Column chooses Right), q = Pr(Row chooses Top).
Step 2. Expected payoff of each decision, for each player.
Step 3. Equate *Row's* expected payoffs to find p; equate *Column's* to find q.

To randomise willingly, a player must be indifferent between the decisions.

Exercise 1.1

Penalty kick – the game

Consider a soccer penalty kick situation where the kicker is equally skilful at kicking to either side, but the goalie is better diving to one side. In particular, the kicker will always score if the goalie dives away from the kick. If the goalie dives to the side of the kick, the kick is always blocked on the goalie's right but is only blocked with probability **one quarter** on the goalie's left.

Represent this as a simple game, in which the goalie earns a payoff of $+1$ for each blocked kick and -1 for each goal, and the kicker earns -1 for each blocked kick and $+1$ for each goal. (Goalie: rows; kicker: columns.)

Exercise 1.1 – step by step

Step 1. Players and actions: goalie (rows) dives L or R; kicker (columns) kicks L or R.

Step 2. Different sides: always a goal $\Rightarrow$ goalie -1, kicker $+1$.

Step 3. Both R: always blocked $\Rightarrow$ goalie $+1$, kicker -1.

Step 4. Both L: blocked with probability $\frac{1}{4}$, goal with probability $\frac{3}{4}$. Expected payoffs:

$$\text{goalie: } \tfrac{1}{4}(+1) + \tfrac{3}{4}(-1) = -\tfrac{1}{2} \qquad \text{kicker: } \tfrac{1}{4}(-1) + \tfrac{3}{4}(+1) = +\tfrac{1}{2}$$

Step 5. The matrix (goalie, kicker):

	Kicker L	Kicker R
Goalie L	$-\frac{1}{2},\ \frac{1}{2}$	$-1,\ 1$
Goalie R	$-1,\ 1$	$1,\ -1$

Check: in every cell the payoffs sum to zero. No pure-strategy equilibrium: whatever is chosen, one of the two wants to switch.

Exercise 1.2

Penalty kick – the equilibrium

Same game as Exercise 1.1. Let p be the probability that the kicker kicks *Left* and q the probability that the goalie dives *Left*.

Determine the equilibrium probabilities used in the Nash equilibrium.

Exercise 1.2 – step by step

Step 1. The goalie must be indifferent; his payoffs depend on p (kicker):

$$\mathbb{E}_G[L] = -\tfrac{1}{2}\,p + (-1)(1-p) = -1 + \tfrac{p}{2} \qquad \mathbb{E}_G[R] = -1\cdot p + 1\cdot(1-p) = 1 - 2p$$

Step 2. Equate: $-1 + \frac{p}{2} = 1 - 2p \;\Rightarrow\; \frac{5}{2}p = 2 \;\Rightarrow\; p = \frac{4}{5}$.

Step 3. The kicker must be indifferent; his payoffs depend on q (goalie):

$$\mathbb{E}_K[L] = \tfrac{1}{2}\,q + 1\cdot(1-q) = 1 - \tfrac{q}{2} \qquad \mathbb{E}_K[R] = 1\cdot q + (-1)(1-q) = 2q - 1$$

Step 4. Equate: $1 - \frac{q}{2} = 2q - 1 \;\Rightarrow\; \frac{5}{2}q = 2 \;\Rightarrow\; q = \frac{4}{5}$.

Step 5. Check: with $p = 0.8$, $\mathbb{E}_G[L] = -0.6 = \mathbb{E}_G[R]$; with $q = 0.8$, $\mathbb{E}_K[L] = 0.6 = \mathbb{E}_K[R]$.

Both choose Left 80% of the time. The goalie dives more often to his *weak* side, and the kicker aims there more often too: each mix is pinned down by the *other* player's payoffs.

PART 2

Battle of the sexes and coordination

Battle of the sexes

Lecture 9 – “The battle of the sexes”, “An experiment”, “Expected payoffs”, “Nash equilibria”

	West	East
West	$2, 1$	$0, 0$
East	$0, 0$	$1, 2$

p = probability that Column chooses Right (East):

$$\mathbb{E}[\text{Top}] = 2\,(1-p) + 0 \cdot p = 2 - 2p$$

$$\mathbb{E}[\text{Bottom}] = 0\,(1-p) + 1 \cdot p = p$$

$$2 - 2p = p \;\Rightarrow\; p = 2/3$$

- Three Nash equilibria: two in pure strategies (upper-left and lower-right corners of the best-response graph) and one mixed, where each player chooses the preferred location with probability $2/3$.
- Experiment, 30 students, random matching: the preferred location is chosen about 67% of the time.
- With the same partner, people tend to take turns.

Exercise 2.1

A two-player game

The Row player can choose between “up” and “down”; the Column player can choose between “left” and “right”.

- If Row chooses up and Column chooses left: Row earns 5, Column earns 2.
- If Row chooses down and Column chooses right: Row earns 3, Column earns 8.
- In all other cases both players earn 0.

(a) Write the payoff table for this game.

(b) Select which category this game belongs to: a. Prisoner’s dilemma b. Matching pennies game c. Battle of the sexes d. Trust game

(c) Select all the Nash equilibria in pure strategies: a. DOWN–RIGHT b. DOWN–LEFT c. UP–LEFT d. UP–RIGHT

Exercise 2.1 – step by step

(a) **Step 1.** (Up, Left) $= (5, 2)$; (Down, Right) $= (3, 8)$; other cells $(0, 0)$.

	Left	Right
Up	$\underline{5}, \underline{2}$	$0, 0$
Down	$0, 0$	$\underline{3}, \underline{8}$

(b) **Step 1.** Both earn something only if they coordinate, but Row prefers (Up, Left), 5 against 3, and Column prefers (Down, Right), 8 against 2 $\Rightarrow$ **c. Battle of the sexes**.

Step 2. Not a prisoner's dilemma: that has a single equilibrium. Not matching pennies: that has no pure-strategy equilibrium. Not a trust game: that is sequential.

(c) **Step 1.** Row: against Left, Up $5 > 0$; against Right, Down $3 > 0$. **Step 2.** Column: against Up, Left $2 > 0$; against Down, Right $8 > 0$.

Step 3. Both payoffs underlined in UP–LEFT and DOWN–RIGHT $\Rightarrow$ **a** and **c**.

Exercise 2.2

Expected profits and mixed strategies

Same game as Exercise 2.1. Let p denote the probability that the Column player chooses "right", and q the probability that the Row player chooses "up".

(a) Complete the following equations for the expected profits of each player's possible choices:

$$\mathbb{E}[\pi_{\text{up}}] = \square\,(1-\square) + \square \cdot \square \qquad \mathbb{E}[\pi_{\text{left}}] = \square \cdot \square + \square\,(1-\square)$$
$$\mathbb{E}[\pi_{\text{down}}] = \square\,(1-\square) + \square \cdot \square \qquad \mathbb{E}[\pi_{\text{right}}] = \square \cdot \square + \square\,(1-\square)$$

(b) The equilibrium in mixed strategy of this game prescribes that the probability that the Column player plays Right is $p = ?$, and the probability that the Row player plays Up is $q = ?$

Exercise 2.2 – step by step

(a) Step 1. Row's payoffs depend on Column: Right with probability p, Left with $1-p$.

$$\mathbb{E}[\pi_{\text{up}}] = 5\,(1-p) + 0\cdot p \qquad \mathbb{E}[\pi_{\text{down}}] = 0\,(1-p) + 3\cdot p$$

Step 2. Column's payoffs depend on Row: Up with probability q, Down with $1-q$.

$$\mathbb{E}[\pi_{\text{left}}] = 2\cdot q + 0\,(1-q) \qquad \mathbb{E}[\pi_{\text{right}}] = 0\cdot q + 8\,(1-q)$$

(b) Step 3. Row indifferent: $5-5p = 3p \;\Rightarrow\; 5 = 8p \;\Rightarrow\; p = \frac{5}{8} = 0.625$.

Step 4. Column indifferent: $2q = 8-8q \;\Rightarrow\; 10q = 8 \;\Rightarrow\; q = \frac{4}{5} = 0.8$.

Step 5. Check: $p = 0.625$: up $= 1.875$, down $= 1.875$. $q = 0.8$: left $= 1.6$, right $= 1.6$.

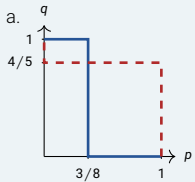
Typical mistake: solving Row's indifference for q. Row's payoffs depend only on p, so Row's indifference pins down p, Column's mix.

Exercise 2.3

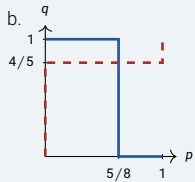
Best-response functions

Same game. Please select the graph that gives the correct representation of the best-response functions (solid: Row's best response, dashed: Column's best response).

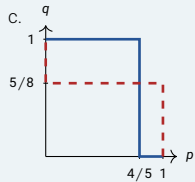
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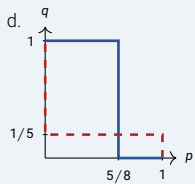
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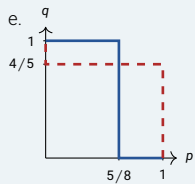
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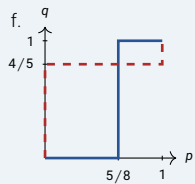
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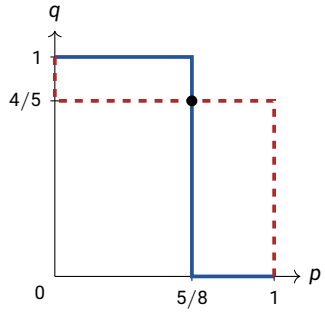
e.



f.



Exercise 2.3 – step by step



Step 1. Row: Up if $5 - 5p > 3p \iff p < 5/8 \Rightarrow q = 1$ for $p < 5/8$, $q = 0$ for $p > 5/8$ (solid line steps *down*).

Step 2. Column: Right if $8 - 8q > 2q \iff q < 4/5$ $\Rightarrow p = 1$ for $q < 4/5$, $p = 0$ for $q > 4/5$.

Step 3. Eliminate: a. threshold $3/8$; b. Column's line the wrong way; c. thresholds swapped; d. q-threshold $1/5$; f. both lines the wrong way $\Rightarrow$ **answer e**.

Step 4. One intersection per equilibrium: $(p, q) = (0, 1)$ Up–Left, $(1, 0)$ Down–Right, $(5/8,\ 4/5)$ mixed.

Exercise 2.4

A coordination game

Consider again the coordination game of Lecture 6.

	Left	Right
Top	3, 3	10, 0
Bottom	0, 10	20, 20

This game has two pure-strategy Nash equilibria. Does it have any equilibrium in mixed strategies? If so, find it.

Exercise 2.4 – step by step

Step 1. p = Pr(Column chooses Right). Row's expected payoffs:

$$\mathbb{E}[\text{Top}] = 3\,(1-p) + 10\,p = 3 + 7p \qquad \mathbb{E}[\text{Bottom}] = 0\,(1-p) + 20\,p = 20p$$

Step 2. Row indifferent: $3 + 7p = 20p \;\Rightarrow\; 13p = 3 \;\Rightarrow\; p = \frac{3}{13} \approx 0.23$.

Step 3. q = Pr(Row chooses Top). Column's expected payoffs:

$$\mathbb{E}[\text{Left}] = 3q + 10\,(1-q) = 10 - 7q \qquad \mathbb{E}[\text{Right}] = 0 \cdot q + 20\,(1-q) = 20 - 20q$$

Step 4. Column indifferent: $10 - 7q = 20 - 20q \;\Rightarrow\; 13q = 10 \;\Rightarrow\; q = \frac{10}{13}$, i.e. Bottom with probability $\frac{3}{13}$.

Step 5. **Yes**: a third equilibrium, in which each player chooses the high-effort action (Bottom, Right) with probability $3/13$.

Note: $3/13$ is the same threshold found for this game in Lecture 7 ("Learning and basin of attraction", actions Low/High): $\Delta = 13p_H - 3 = 0$.

PART 3

Payoff imbalances and noisy best responses

Payoff imbalances

Row’s Top/Left payoff	Row indifferent at	Column indifferent at	prediction
72 (balanced)	$p = 1/2$	$q = 1/2$	Top 50%
360	$p = 0.9$	$q = 1/2$	Top 50%
40	$p = 0.1$	$q = 1/2$	Top 50%

- In a Nash equilibrium, Row mixes to keep *Column* indifferent: Row’s own payoffs change p (Column’s mix), not q (Row’s mix).
- Data: Top rises from 36% (payoff 40) to 67% (payoff 360); Column’s Right from 24% to 74%. Behaviour reacts to a player’s **own** payoffs.

Noisy best responses and QRE

Lecture 10 – “Noisy Best Responses”, “Varying the amount of noise”, “Quantal Response Equilibrium”

Noisy best response: the probability of a decision rises *smoothly* with its expected payoff advantage, instead of jumping from 0 to 1.

$$\Pr(R) = \frac{\exp(\pi_R \lambda)}{\exp(\pi_R \lambda) + \exp(\pi_L \lambda)} \qquad \Pr(T) = \frac{\exp(\pi_T \lambda)}{\exp(\pi_T \lambda) + \exp(\pi_B \lambda)}$$

- Less noise: the curves get closer to the step-shaped best responses.
- **QRE**: choice probabilities match the beliefs that generate them (intersection of the noisy response curves).
- Symmetric matching pennies: the intersection stays at the centre, QRE = NE. With payoff imbalances, the QRE moves in the direction of the own-payoff effect observed in the data.

Exercise 3.1

A game

Consider a game with the following payoff matrix.

	Left	Right
Top	14, 5	5, 11
Bottom	5, 9	8, 5

Identify the pure-strategy equilibria of this game.

a. (Top, Left), with an outcome of (14, 5)
b. (Top, Right), with an outcome of (5, 11)
c. (Bottom, Left), with an outcome of (5, 9)
d. (Bottom, Right), with an outcome of (8, 5)
e. None of the above, the game does not have any equilibrium in pure strategies

Exercise 3.1 – step by step

Step 1. Row's best responses: against Left, Top $\mathbf{14} > \mathbf{5}$; against Right, Bottom $\mathbf{8} > \mathbf{5}$.

Step 2. Column's best responses: against Top, Right $\mathbf{11} > \mathbf{5}$; against Bottom, Left $\mathbf{9} > \mathbf{5}$.

	Left	Right
Top	$\underline{\mathbf{14}}, 5$	$5, \underline{\mathbf{11}}$
Bottom	$5, \underline{\mathbf{9}}$	$\underline{\mathbf{8}}, 5$

Step 3. No cell has both payoffs underlined $\Rightarrow$ **e**.

Step 4. Follow the arrows: (Top, Left) $\rightarrow$ Column switches to Right $\rightarrow$ Row switches to Bottom $\rightarrow$ Column switches to Left $\rightarrow$ Row switches to Top. A cycle, as in matching pennies: the only equilibrium is in mixed strategies.

Exercise 3.2

Row's expected payoffs

Same game as Exercise 3.1. Let p denote the probability that the Column player chooses "Right", and q the probability that the Row player chooses "Top".

Find the expression for the Row player's expected payoffs, one for each of Row's decisions.

Exercise 3.2 – step by step

Step 1. Row's payoffs depend on what Column does: Left with probability $1 - p$, Right with probability p.

Step 2. Read Row's payoffs from the Top row: 14 against Left, 5 against Right.

$$\mathbb{E}_R[\text{Top}] = 14\,(1 - p) + 5\,p = 14 - 9p$$

Step 3. Read Row's payoffs from the Bottom row: 5 against Left, 8 against Right.

$$\mathbb{E}_R[\text{Bottom}] = 5\,(1 - p) + 8\,p = 5 + 3p$$

Typical mistake: using Column's payoffs (the second number in each cell), or writing Row's payoffs as functions of q.

Exercise 3.3

Column's expected payoffs

Same game. Let p denote the probability that the Column player chooses "Right", and q the probability that the Row player chooses "Top".

Find the expression for the Column player's expected payoffs, one for each of Column's decisions.

Exercise 3.3 – step by step

Step 1. Column's payoffs depend on what Row does: Top with probability q, Bottom with probability $1 - q$.

Step 2. Column's payoffs in the Left column: 5 against Top, 9 against Bottom.

$$\mathbb{E}_C[\text{Left}] = 5q + 9(1 - q) = 9 - 4q$$

Step 3. Column's payoffs in the Right column: 11 against Top, 5 against Bottom.

$$\mathbb{E}_C[\text{Right}] = 11q + 5(1 - q) = 5 + 6q$$

Exercise 3.4

The mixed equilibrium

Same game. Let p denote the probability that the Column player chooses “Right”, and q the probability that the Row player chooses “Top”.

Compute the equilibrium in mixed strategies, if it exists.

Exercise 3.4 – step by step

Step 1. Row indifferent (Exercise 3.2):

$$14 - 9p = 5 + 3p \;\Rightarrow\; 9 = 12p \;\Rightarrow\; p = \tfrac{3}{4}$$

Step 2. Column indifferent (Exercise 3.3):

$$9 - 4q = 5 + 6q \;\Rightarrow\; 4 = 10q \;\Rightarrow\; q = \tfrac{2}{5}$$

Step 3. Check: $p = \frac{3}{4}$: Top $= 14 - 6.75 = 7.25$, Bottom $= 5 + 2.25 = 7.25$. $q = \frac{2}{5}$: Left $= 9 - 1.6 = 7.4$, Right $= 5 + 2.4 = 7.4$.

Step 4. Equilibrium: Column plays Right with probability $3/4$, Row plays Top with probability $2/5$.

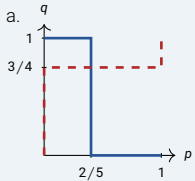
Notice: Row's large payoff of 14 in (Top, Left) does *not* make Row play Top more often in the Nash equilibrium ($q = 0.4$ comes from Column's payoffs). It makes *Column* play Right more often, to keep Row indifferent.

Exercise 3.5

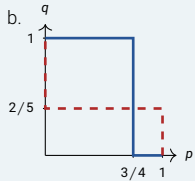
Best-response functions

Same game. Please select the graph that gives the correct representation of the best-response functions relative to the game under consideration (solid: Row, dashed: Column).

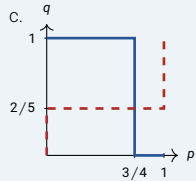
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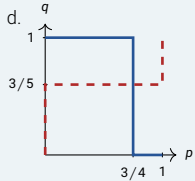
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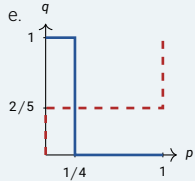
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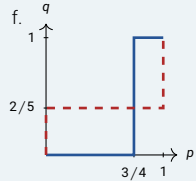
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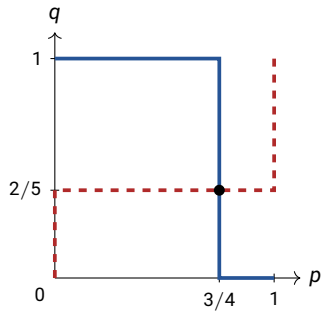
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Exercise 3.5 – step by step



Step 1. Row: Top if $14 - 9p > 5 + 3p \iff p < 3/4$ $\Rightarrow q = 1$ for $p < 3/4$, $q = 0$ for $p > 3/4$.

Step 2. Column: Right if $5 + 6q > 9 - 4q \iff q > 2/5 \Rightarrow p = 0$ for $q < 2/5$, $p = 1$ for $q > 2/5$. Unlike the battle of the sexes, Column's line goes *up* at the threshold.

Step 3. Eliminate: a. thresholds swapped; b. Column's line the wrong way; d. q-threshold $3/5$; e. p-threshold $1/4$; f. Row's line the wrong way $\Rightarrow$ **answer c**.

Step 4. A single intersection, $(p, q) = (3/4,\ 2/5)$: the unique, mixed, equilibrium. No corner is on both lines.

Exercise 3.6

Noisy best responses

Explain what a “noisy best response” is and discuss whether the equilibrium with noisy best responses (quantal response equilibrium) would be the same as the Nash equilibrium or not.

Exercise 3.6 – step by step

Step 1. **Best response**: a step function. Row plays Top for sure as soon as $p < 3/4$, even if Top is better by a cent.

Step 2. **Noisy best response**: the probability of Top falls smoothly as p rises. It is close to 1 when Top is much better, close to 0 when it is much worse, and near $1/2$ when the two decisions give similar expected payoffs. The step becomes an S-shaped curve; less noise, sharper corners.

Step 3. **QRE**: the point where the two noisy response curves cross, so that choice probabilities are consistent with the beliefs that generate them.

Step 4. **Same as NE?** In symmetric matching pennies, yes: the curves still cross at the centre. Here, with a payoff imbalance, in general **no**: Row's large payoff in (Top, Left) makes Top more attractive for a wide range of beliefs, so the noisy curves cross at a point where Row plays Top more often than the Nash prediction $q = 0.4$.

This is the own-payoff effect of Lecture 10: when Row's Top/Left payoff rose from 40 to 360, Top choices rose from 36% to 67%, although Nash predicts 50% in both cases.

PART 4

The playing-cards exercise

Exercise 4.1

Playing cards – Lecture 10

The row player has a red 10 (hearts or diamonds) and a black 2 (clubs or spades). The column player has a black 6 and a red 2. They have to choose which card to play, simultaneously.

- If the colours of the cards do not match (red and black), each earns nothing.
- If the colours match, earnings for each person are equal to the number on the card that the person played.

(a) Write out the payoff matrix for this game.

(b) Find all equilibria in non-random strategies.

Exercise 4.1 – step by step

(a) **Step 1.** Go through the four combinations:

- red 10 against black 6: colours differ $\Rightarrow (0, 0)$
- red 10 against red 2: match $\Rightarrow$ each gets their own card, $(10, 2)$
- black 2 against black 6: match $\Rightarrow (2, 6)$
- black 2 against red 2: colours differ $\Rightarrow (0, 0)$

	Black 6	Red 2
Red 10	$0, 0$	$\underline{10}, \underline{2}$
Black 2	$\underline{2}, \underline{6}$	$0, 0$

(b) **Step 1.** Row: against Black 6, Black 2 ($2 > 0$); against Red 2, Red 10 ($10 > 0$). **Step 2.** Column: against Red 10, Red 2 ($2 > 0$); against Black 2, Black 6 ($6 > 0$).

Step 3. Two equilibria: **(Red 10, Red 2)**, payoffs $(10, 2)$, and **(Black 2, Black 6)**, payoffs $(2, 6)$. Both want to match colours, but each prefers a different match: a battle of the sexes.

Exercise 4.2

Playing cards – randomised strategies

Same game as Exercise 4.1.

(a) Find expressions for the Row player's expected payoffs, one for each decision, where p denotes the probability that the other player chooses their high-number card.

(b) Find expressions for the Column player's expected payoffs for each decision, black 6 or red 2, where q denotes the probability that Row plays the red 10.

(c) Find the equilibrium with randomised strategies.

Exercise 4.2 – step by step

(a) Step 1. Column's high card is the black 6: played with probability p; red 2 with probability $1-p$.

$$\mathbb{E}_R[\text{Red } 10] = 0\cdot p + 10\,(1-p) = 10 - 10p \qquad \mathbb{E}_R[\text{Black } 2] = 2\,p + 0\,(1-p) = 2p$$

(b) Step 1. Row plays the red 10 with probability q, the black 2 with $1-q$.

$$\mathbb{E}_C[\text{Black } 6] = 0\cdot q + 6\,(1-q) = 6 - 6q \qquad \mathbb{E}_C[\text{Red } 2] = 2\,q + 0\,(1-q) = 2q$$

(c) Step 1. Row indifferent: $10 - 10p = 2p \;\Rightarrow\; 12p = 10 \;\Rightarrow\; p = \frac{5}{6}$.

Step 2. Column indifferent: $6 - 6q = 2q \;\Rightarrow\; 8q = 6 \;\Rightarrow\; q = \frac{3}{4}$.

Step 3. Mixed equilibrium: Column plays the black 6 with probability $5/6$, Row plays the red 10 with probability $3/4$. Check: $10 - \frac{50}{6} = \frac{10}{6} = 2\cdot\frac{5}{6}$; $\;6 - 4.5 = 1.5 = 2\cdot\frac{3}{4}$.

Exercise 4.3

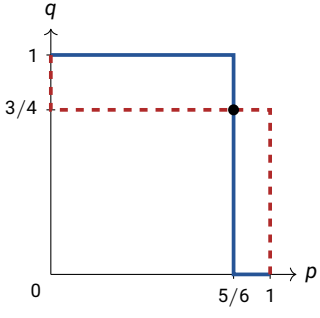
Playing cards – graph

Same game.

(a) Draw the best-reply functions of the two players, in a graph.

(b) How would the graph change if we allowed for some randomness in choices, as modelled in the Quantal Response Equilibrium?

Exercise 4.3 – step by step



p = Pr(black 6), q = Pr(red 10)

(a) **Step 1.** Row: red 10 if $10 - 10p > 2p \iff p < 5/6$ $\Rightarrow q = 1$ for $p < 5/6$, $q = 0$ above.

Step 2. Column: black 6 if $6 - 6q > 2q \iff q < 3/4$ $\Rightarrow p = 1$ for $q < 3/4$, $p = 0$ above.

Step 3. Three intersections: $(0, 1)$ = (Red 10, Red 2); $(1, 0)$ = (Black 2, Black 6); $(5/6,\ 3/4)$ mixed.

(b) **Step 1.** The steps become smooth S-shaped curves: choices respond gradually to expected payoff differences.

Step 2. The corners are rounded, so the intersections are no longer exactly at $(0, 1)$ and $(1, 0)$: players never choose a card with certainty. Less noise, curves closer to the steps.

PART 5

Oral questions

Sample oral questions I

Randomised strategies

1. Why can it be optimal to be unpredictable? Give an example of a game with no equilibrium in pure strategies.
2. Why must a player be indifferent between her decisions in order to randomise? How is the mixed equilibrium computed?
3. Describe the battle-of-the-sexes experiment of Lecture 9. How does behaviour compare with the equilibria of the game?

Noisy best responses

1. What does Nash equilibrium predict when one player's own payoff in a matching pennies game is increased? What do experiments show?
2. What is a noisy best response, and what is a quantal response equilibrium?

Sample oral questions II

Base-rate fallacy

1. To discuss the base-rate fallacy, we have considered a famous experiment in which respondents had to assess the probability that an individual (named Dick) was a Lawyer or an Engineer. Describe the experiment.
2. What is the correct reasoning according to the Bayes' rule?
3. What is instead the answer given by most of the respondents?
4. What is the base rate fallacy?